

MPPT BASED CONTROL OF SEPIC CONVERTER USING FIREFLY ALGORITHM FOR SOLAR PV SYSTEM UNDER PARTIAL SHADED CONDITIONS

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Abstract: This paper presents a MPPT based control of SEPIC converter using Firefly algorithm for Solar PV system under Partial Shaded Conditions. A novel Firefly Algorithm(FA) is proposed to track the maximum power from solar panel. The proposed Firefly algorithm eliminates the multiple peaks, reduce the tracking time and improve the conversion efficiency of the converter. The results of the performance parameters at different irradiation and Partial Shaded Conditions are compared and analyzed for Perturb and Observe (P&O), Particle Swarm Optimization (PSO) and Firefly algorithms. The results of the proposed algorithm show better results compared with other algorithms.

Keywords: Firefly algorithm(FA), Maximum power Point Tracking(MPPT), Partial Shaded Conditions(PSCs), Photovoltaic(PV) systems, Perturb and Observe(P&O) method

1. INTRODUCTION

WITH growing world energy demand and soaring prices of fossil fuels together with concern about environmental issues have generated enormous interest in the utilization of renewable energy sources. The photovoltaic (PV) power generation has seen a rapid growth in the last few years leading to extensive use of solar energy; a PV system has the advantages of low maintenance cost, absence of moving or rotating parts and freedom from environmental pollution [5], [6]. Many countries provide generous financial schemes such as feed-in tariff [7], subsidized policies [8], etc., leading to rapid growth of PV power generation systems. The objective of MPPT is to extract maximum power generated by the PV systems under varying condition of temperature and solar insolation. A major challenge in PV systems is to tackle its nonlinear current-voltage ($I-V$) characteristics,

leading to a unique maximum power point (MPP) on its power-voltage ($P-V$) characteristic curve. The process of MPPT is complicated by the fact that the $P-V$ curves vary largely with solar insolation and temperature.

Generally, the PV panels are connected in series and parallel so as to meet the load power demand. [3], [4]. When climatic conditions vary, the MPP of the PV system also changes its position and several methods have been presented for tracking the MPP and are available in [13] [14]. These methods include perturb and observe (P&O), incremental conductance [15], short circuit current [16], open circuit voltage [17], load current/load voltage maximization technique [19], fuzzy control [18], neural network-based schemes [20], etc. A detailed comparison of various methods for tracking MPP in PV systems is extensively discussed in [21] [22]. The tracking methods discussed in these papers are effective and time tested under uniform solar insolation, where there is only one MPP in the $P-V$ curve of the PV system for a given temperature and insolation. In large PV systems, partially shaded condition (PSC) occurs wherein PV modules receive different solar insolation due to shadow of building, moving clouds, and other neighbouring objects. The output power of the PV array decreases largely due to PSC and the quantum of power lost depend on system configuration, shading pattern and the bypass diodes incorporated in the PV modules. The effect of PSC on PV system has been investigated in several publications. The immediate effect of PSC is that the resulting PV characteristic curve becomes complex with multiple peaks. Conventional methods of tracking MPP are based on "hill climbing" technique and these methods are not effective in reaching the global optima, when the PV system under PSC exhibits multiple peaks; rather most of the conventional methods may converge to local MPP leading to power loss. These methods are calculating the MPP using different ways under the same fundamental concept. In these algorithms the first

local maximum point (LMP) is reached means stop maximum point searching. This is the main drawback of these algorithms [10]. To maximize the efficiency of solar module by tracking global maximum point (GMP) from solar panel.

Basically two approaches are used to find GMP. The first method is using the hard ware i.e. use of multilevel inverter but they produce complexity and costly. The second method is to track the GMP with help of control algorithms and the paper also discusses the control algorithms. In this paper gives the P&O, PSO and FA methods. Starting the P&O method employed to identify the near local maximum point. From that point the FA used to search the GMP. The advantage of this method tracks the maximum power point in very less time and convergence speed is greatly increased.

II. PERTURB & OBSERVE METHOD

P&O method is the most widely algorithm used to track the maximum power of PV because it use only a few parameters and the simple feedback structure. For practical implementation, the P&O method can be selected because had a good performance in tracking MPPT and it always made the operating point oscillate near the MPP. The P&O controller had also better response time, less oscillation and much more accurate tracking at each step. The algorithm takes the value of voltage and current of solar panel. Power is calculated with help of voltage and current[7]. The value of voltage and current are denoted by k^{th} instant [1] [2]. Next value of voltage and currents are mentioned $k+1$ th instant and the next instants are measured from the previous instants.

The change of power is interrupted indirectly will disturb voltage on PV terminal for the entire cycle of maximum power point tracking. By keeping the output voltage of the PV terminal, so the maximum power point will be achieved. The entire process shown as flow chart in figure 1.

The main drawback of the method is first local maximum point is reached means the searching will be stop in that place. So the global maximum point is not attained and no guarantee about the global maximum power point. By keeping the output voltage of the PV terminal, so the maximum power point will be achieved. P&O method scheme can be explained by the following mathematical equation.

$$\text{On the condition of the voltage source} \quad \frac{\delta p}{\delta v} > 0, v_{\text{ref}} = v_{\text{ref}} + \Delta v_{\text{ref}} \quad (1)$$

On the condition of the current source

$$\frac{\delta p}{\delta v} < 0, v_{\text{ref}} = v_{\text{ref}} - \Delta v_{\text{ref}} \quad (2)$$

On the condition of the power point

$$\frac{\delta p}{\delta v} < 0, v_{\text{ref}} = v_{\text{ref}} - \Delta v_{\text{ref}} \quad (3)$$

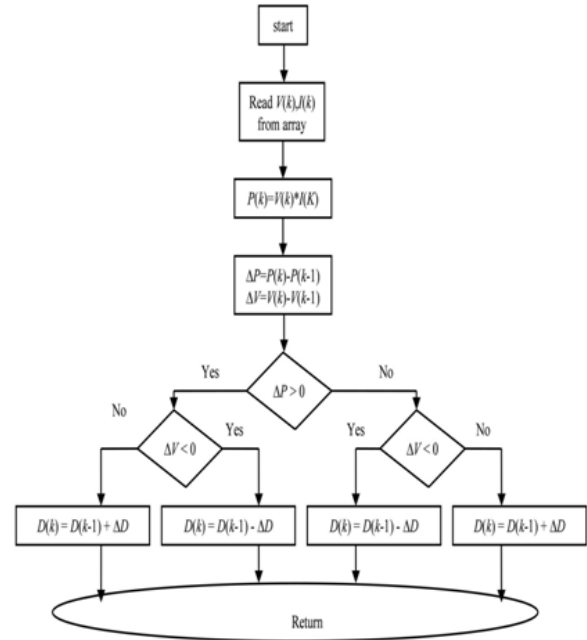


Fig.1. Flow chart for P&O method

III. PARTICLE SWARM OPTIMIZATION

A population based optimization technique inspired by social behavior of bird flocking or fish schooling, PSO consists of a swarm of particles. Each particle resides at a position in the search space [8] [9]. The fitness of each particle represents the quality of its position. The particles fly over the search space with a certain velocity [10]. The velocity (both direction and speed) of each particle is dependent on its own best position found so far and the best possible solution. Eventually the swarm will converge to optimal positions by using (1) and (2) equation.

$$V_i(k+1) = wv_i(k) + c_1r_1.(p_{\text{best}}x_i(k)) + c_2r_2.(g_{\text{best}} - x_i(k)) \quad (4)$$

$$x_i(k+1) = x_i(k) + v_i(k+1) \quad i=1,2,\dots,N \quad (5)$$

where x_i is the position of particle i ; v_i is the velocity of particle i ; k denotes the iteration number; w is the inertia weight; r_1 and r_2 are random variables uniformly distributed within $[0,1]$; and c_1 , c_2 are the cognitive and social coefficient, respectively. The variable $p_{\text{best},i}$ is used to store the best position that

the i th particle has found so far, and gbest is used to store the best position of all the particles. Fig. 2 shows the flowchart of the PSO based MPPT technique.:

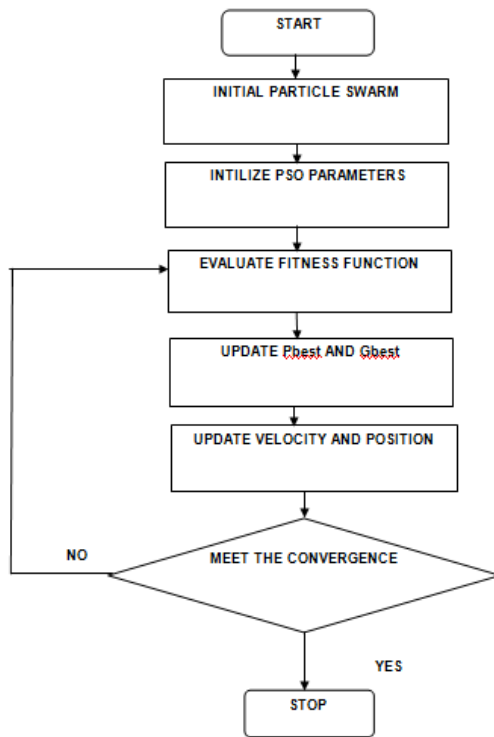


Fig.2. Flow chart for Particle Swarm Optimization

Input: Initialize the algorithm parameters (c_1 , c_2 , w , v_{max} , $Swarm_Size$, Max_Iter , r_1 , r_2).

Output: The optimization having the highest fitness as found by PSO.

Step 1: Set the number of particles and searching parameters along with the limit for position and velocity.

Step 2: Randomly initialize Position and velocity of each particle.

Step 3: Compute the fitness value of each particle.

Step 4: The particle having the best fitness value is set as Gbest (Global Best).

Step 5: Update the position and velocity of each particle with respect to the Gbest.

Step 6: Repeat Step 3 & 4 till the optimum solution is reached.

Step 7: Gbest at the end of the last iteration gives the optimized value.

Step 8: Compute the Duty-cycle.

As the particles approach the MPP, they get closer to the Gbest position. Correspondingly, the Pbest factor and Gbest factor in velocity term moves towards zero. Eventually a zero velocity is achieved and the duty cycle remains almost unchanged. Under this condition, the PV system reaches at MPP. The PSO algorithm will not be trapped in a local maximum because some particles would always look the better global position in sequential iteration.

Such feature is very useful to track MPP in partial shading condition. The advantage that lies with PSO method in finding an optimal duty cycle are (1) PSO search starts with random initial guess, (2) possibility to explore the search region with continuous updation of duty cycles, (3) reduction in steady state oscillations around MPP.

IV. FIRE FLY ALGORITHM

The FA is a population-based optimization and is introduced by Yang [11], [12]. The fire fly algorithm is type of soft computing techniques. This optimization algorithm is inspired by the movement of lightning bugs-commonly known as fireflies[23]. The flashing light of fireflies is an amazing sight in the summer sky in the tropical and temperate regions. Two fundamental functions of such flashes are to attract mating partners and to attract potential prey. In addition, flashing may also serve as a protective warning mechanism. The rhythmic flash, the rate of flashing and the amount of time form part of the signal system that brings both sexes together.

The statistical analysis of the comparison clearly indicates that FA is potentially more powerful in finding global optima with least computing time.

For simplicity in describing Firefly Algorithm, the following three assumptions are made:

1. All fireflies are unisex so that one firefly will be attracted to other fireflies regardless of their sex.
2. The attractiveness between two fireflies is proportional to relative brightness and the less bright one will move toward the brighter one. If there is no brighter one in a firefly colony, each one will move randomly.
3. The brightness of a firefly is affected or determined by the landscape of the objective function. For a maximization problem, the brightness can simply be proportional to the value of the objective function. Let i and j be two fireflies positioned at X_p and X_q respectively. Let the distance between these two fireflies is denoted as r_{pq} a single dimensional space, we can write

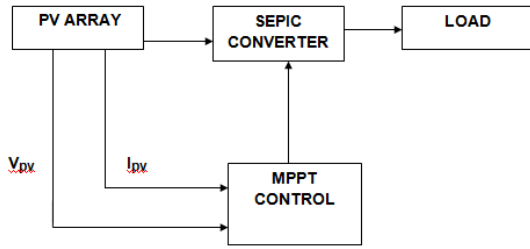


Fig.3. Block diagram of FA-based MPPT scheme.

TABLE I
PARAMETERS OF SEPIC CONVERTER

Switching frequency f_s	20kHz
Capacitor C1	200 μ H
Capacitor C2	100 μ H
Inductor L1	280 μ H
Inductor L2	150 μ H

$$r_{pq} = ||x_p - x_q|| \quad (6)$$

The degree of attractiveness, β is a function of distance between two fireflies and is given by

$$\beta(\gamma) = \beta_0 e^{-\gamma(r_{pq})^n}, \quad n \geq 1 \quad (7)$$

In the aforementioned equation, γ , which controls the decrease of light intensity, is termed as absorption coefficient and is between 0 and 10 and $n=2$. The symbol β_0 is initial attractiveness and is chosen as 1, such that the brightest firefly strongly determines the position of other fireflies in its neighbourhood.

Assuming that the brightness of firefly p is less than that of q , the new position of firefly p is given by the following equation:

$$x_p^{t+1} = x_p^t + \beta(\gamma)(x_p - x_q) \alpha \left(\text{rand} \left(-\frac{1}{2} \right) \right) \quad (8)$$

Here, random movement factor α is constant throughout the program and falls in the range [0, 1] and rand is a random number uniformly distributed between 0 and 1 for each movement of firefly. Large amount of α makes the movement to explore the solution through the distant search space and the smaller α tends to facilitate local search.

The flowchart of the FA-based MPPT scheme is given in Fig. 4. For a population of fireflies, i.e., duty ratios, the digital controller measures V_{pv} and I_{pv} and computes the output power. The steps of FA algorithm toward MPPT are described

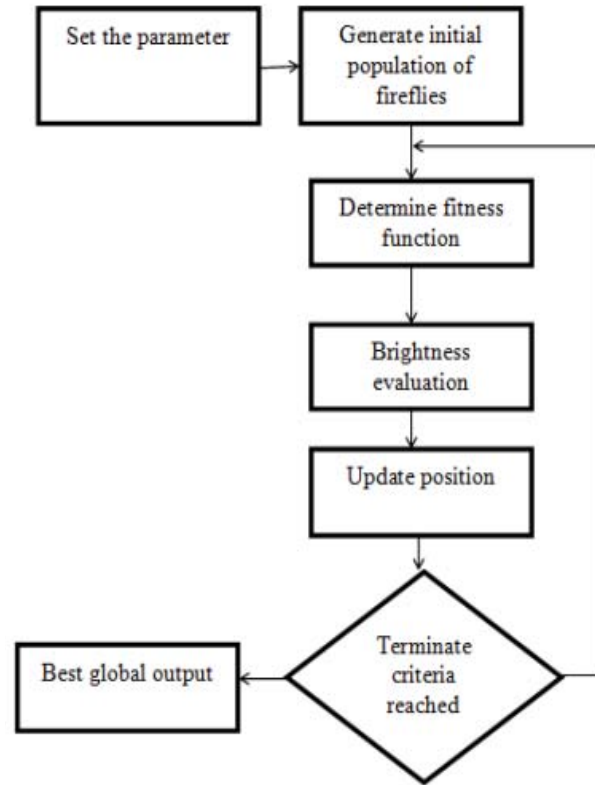


Fig.4. Flowchart for Firefly algorithm

STEP 1: **parameter setting**, Fix the constants of the FA, population size N , and the termination criterion. In this algorithm, the position of the firefly is taken as a duty cycle of the dc-dc converter. The brightness of each firefly is taken as generated power of the PV system, corresponding to the position of this firefly.

STEP 2: **Initialization of Fireflies**, In this step, the fireflies are positioned in the allowable solution space between $d_{\min}$ to $d_{\max}$ where $d_{\min}$ and $d_{\max}$ represent the minimum and maximum values of the duty ratio of the dc-dc converter.

STEP 3: **Brightness Evaluation**, in this step, the dc-dc converter is operated corresponding to the position of each firefly (i.e., duty ratio) sequentially

STEP 4: **Update the Position of Fireflies**, The firefly with maximum brightness remains in its position and the remaining fireflies update their position based on.

STEP 5: **Terminate the program**, if the termination criterion is reached, else go to step 3.

STEP 6: **Reinitiate the FA**, if the solar insolation changes, which is detected by the digital controller by sensing the change in the power output.

To start the algorithm, the fireflies are placed in random locations After a firefly has been moved towards all brighter fireflies, its brightness is updated by evaluating the objective function in the new position.

The new evaluation is also compared against the best found so far. If it is better, that position becomes the new best. In this way, if a firefly passes through a location better than any other found, but for whatever reason it or any other firefly does not end up there by the last iteration of the algorithm, that best location is still recorded.

After re-evaluation, the given firefly is done with its update, and the algorithm moves on to the next firefly. This continues for every firefly, and then one generation of the algorithm

V.SIMULATION RESULTS

The simulation of PV array and MPPT with converters is done using MATLAB/Simulink software. The simulation results are discussed in this Chapter. MATLAB is now a full featured technical computing environment. The simulink is a tool for modelling, simulating and analyzing multi domain dynamic systems. Its primary interface is a graphical block diagramming tool and set of block libraries. By using these library blocks, the design of the SEPIC converter and MPPT algorithms are also modelled based on the flowchart.

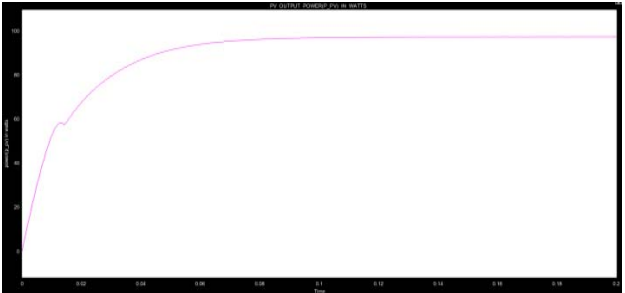
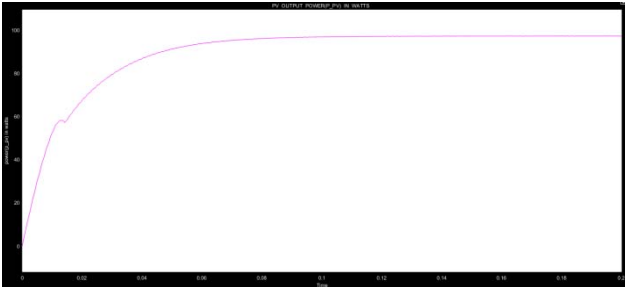
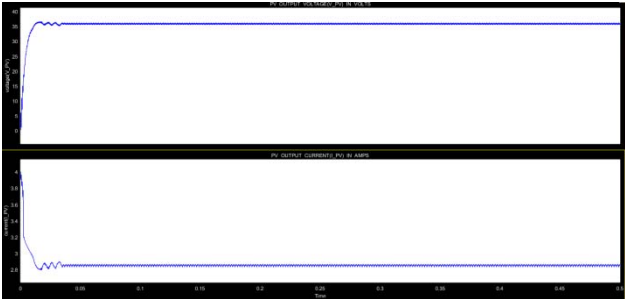
The MPPT consists of three algorithms named as Perturb and Observe, PSO and Fire Fly Algorithm. In this MPPT the above three algorithms are combined to obtain maximum power. In simulation the output voltage, output current, output power of SEPIC converter are represented in scope.

The results and performance parameters at different irradianations and Partial Shaded Conditions are compared and analyzed for P&O, PSO and FA algorithms. The results of the proposed algorithm show better results compared with other algorithms.

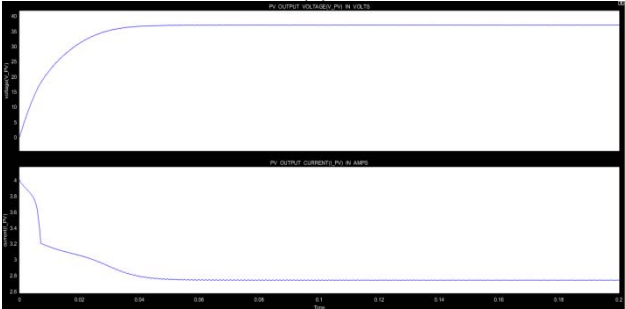
For PV array the desired operating point is Maximum Power Point. In SEPIC converter the values of source voltage Vs and source current, Is are chosen such that Vmp and Imp as (17.5V, 3.5A) respectively. In order to minimum ripples in the output voltage and input current, its assumed that 1% of the output voltage as the ripple voltage V. The switching frequency of the switch is chosen as 20 KHZ. By using the above datas, the SEPIC converter circuit components are calculated.

TABLE I
PARAMETERS OF PV MODULE

Maximum power(p_{max})	120W
Open circuit voltage(v_{oc})	20V
Short circuit current(I_{sc})	4A
Voltage at maximum power(v_{mp})	17.5V
Current at maximum power(I_{mp})	3.5A



(a)



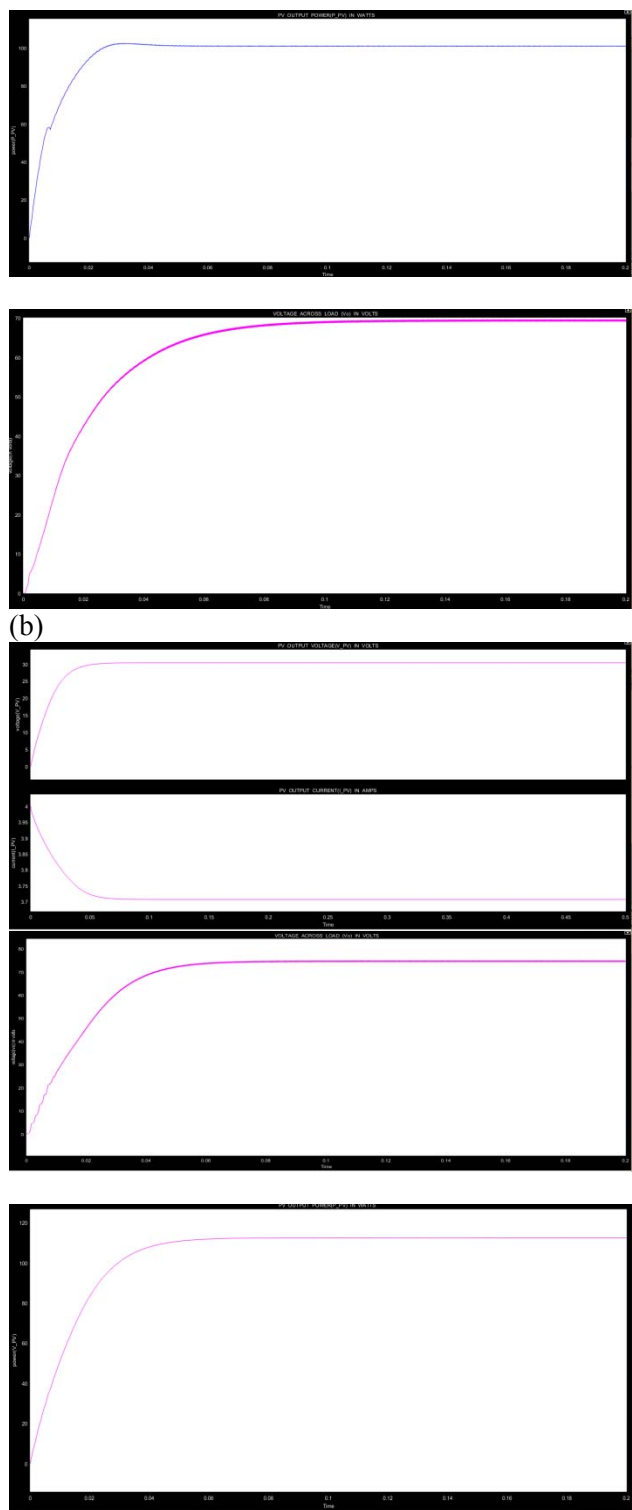


Fig. 5. Simulation results of SEPIC converter with (a)P&O,(b) PSO, and (c) Firefly algorithm under partial shading conditions.

VI. COMPARISON OF ALGORITHMS

The main part of the this report is analysis of SEPIC converter with P&O, PSO and Firefly

algorithms, by comparison of the results evaluated to track the Maximum Power in solar PV system. The SEPIC converter is simulated with P&O, PSO and Firefly algorithms under Partial Shaded Conditions. The simulation result for the output power for SEPIC converter with algorithms is shown and this output power for variable solar irradiance 1000W/m², 800W/m², and 600W/m² at constant temperature 25⁰C. The simulation of SEPIC converter with Firefly algorithm tracks more power under Partial Shaded Condition.

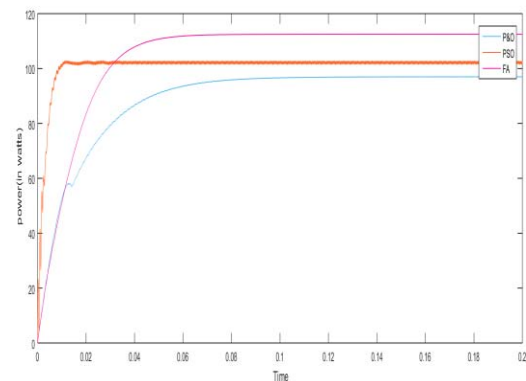


Fig 6: Simulation results of SEPIC converter with P&O, PSO, and Firefly algorithms under partial shaded conditions

The SEPIC results in non-inverted output. The coupling capacitor offers isolation between output side and input side which is unique to SEPIC converter only. In SEPIC converter, MPP tracking can be done with ease because of low input ripple current. In addition to that, SEPIC converter is more efficient than buck-boost converter. The SEPIC converter is easily settled down with irradiance changes. The results with P&O and PSO algorithms to track the local maximum power point, under Partial shaded conditions and the resulting PV characteristic curve becomes complex with multiple peaks. Maximum Power Point tracking time is also high. The Firefly algorithm reduces the complexity of finding GMP, reduce time convergence, and improve dynamic performance. And it also be reduces the multiple peaks.

The results of the proposed algorithm show better results compared with other algorithms.

VII. CONCLUSION

The performance analysis of MPPT based control of SEPIC converter to track the Global maximum Power under Partially Shaded Conditions is presented The SEPIC converter with P&O and PSO and Firefly algorithms are simulated using

MATLAB/simulink. The results of the performance parameters at different irradiation and Partial Shaded Conditions are compared and analyzed. The Firefly algorithm (FA) tracks the maximum power point in very less time and also the output remains constant. FA reduces the complexity of finding Global Maximum Power, and reduce the convergence, improve the conversion efficiency of the converter. In comparison between P&O, PSO and Firefly algorithms, the results of the proposed algorithm show better performance compared with other algorithms

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